BT5340: Protein Folding and Stability

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Assignment 2 Report



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Abstract

This study (Assignment) conducted by us aims to look further into the folding kinetics and thermodynamics of a protein (Wild type and its mutants). We use Chevron analysis and the protein engineering approach to understand the behaviour of our protein. The data being provided to us contains information on the type of mutation , folding rates as well as the m_f and m_u which correspond to the sensitivity of folding/unfolding barrier to the denaturant. For our system GuHCl is used as the denaturant. Finally we extend the protein engineering approach to a three state ($U \rightleftharpoons I \rightleftharpoons F$) system and finally describing the thermodynamics and kinetics for it.

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Introduction

The protein engineering approach was first developed by Fersht and his team [1] in which they used site-directed mutagenesis to understand what the transition state looks like during protein folding process.

In this method small changes (in most cases point mutations are made to the protein) and we observe how these changes affect folding. The ϕ-value tells us how much of the effect of a mutation is already present in the transition state compared to the fully folded protein. The Mathematics behind the formulation of the ϕ value can be explained using the formula :

$$\phi = \frac{\Delta\Delta G^{\ddagger}}{\Delta\Delta G_{\mathrm{eq}}} = \frac{\ln(k_{\mathrm{f}}^{\mathrm{mut}}/k_{\mathrm{f}}^{\mathrm{WT}})}{\ln(K_{\mathrm{eq}}^{\mathrm{mut}}/K_{\mathrm{eq}}^{\mathrm{WT}})} \tag{1}$$

This equation gives us a lot of information regarding the folding process of the protein of interest:

when $\phi = 0$, it means the transition state is fully unfolded-like (the mutation has 0 impact), and when $\phi = 1$ it means that the effects of the mutation is completely felt. Its important to note that Residues with high ϕ-values are considered part of the *folding nucleus* the core of structure that forms in the rate-limiting step. It suggests that mutations in this region affects the stability of both the transition state as well as the final folded state and higher values would indicate that the protein has reached its final conformation during the transition state itself.

Another important thing is that measuring kinetics at different denaturant concentrations gives something called a *chevron plot*. The two sides (arms) of this plot basically tell about the energy barriers for folding and unfolding. Also, one can directly figure out how stable the protein is at equilibrium just by taking the ratio of the folding and unfolding rate constants.

Using this as our background we can move to the next section where we delve deeper into some of the theory and math behind all these concepts

Theory and Methods

Two-state kinetics and the chevron plot

For a two-state system $\mathrm{U} \rightleftharpoons \mathrm{F}$, the folding and unfolding rate constants vary linearly in the exponential with denaturant concentration $[D]$ like :

$$k_{\mathrm{f}}([D]) = k_{\mathrm{f}}^{\mathrm{w}} \exp\left(\frac{m_f\,[D]}{RT}\right), \quad m_f < 0 \tag{2}$$

$$k_{\mathrm{u}}([D]) = k_{\mathrm{u}}^{\mathrm{w}} \exp\left(\frac{m_u\,[D]}{RT}\right), \quad m_u > 0 \tag{3}$$

where m_f and m_u (units $\mathrm{kJ\,mol^{-1}\,M^{-1}}$) are the same kinetic m-values previously seen in the Introduction.

The observed rate is given by the equation :

$$k_{\mathrm{obs}}([D]) = k_{\mathrm{f}}([D]) + k_{\mathrm{u}}([D]) \tag{4}$$

On a semi-log plot $\log_{10}(k_{\mathrm{obs}})$ vs. $[D]$, results in a V shaped chevron plot that has a folding arm (negative slope $\propto m_f$) and an unfolding arm (positive slope $\propto m_u$). The minimum value for K_{obs} occurs where:

$$\frac{\mathrm{d}k_{\mathrm{obs}}}{\mathrm{d}[D]} = 0 \implies \frac{k_{\mathrm{f}}([D])}{k_{\mathrm{u}}([D])} = \frac{m_u}{|m_f|} \implies [D]_{\mathrm{min}} = \frac{RT\,\ln\left(\dfrac{m_u\,k_{\mathrm{u}}^{\mathrm{w}}}{|m_f|\,k_{\mathrm{f}}^{\mathrm{w}}}\right)}{m_f - m_u} \tag{5}$$

Keep in mind that $[D]_{\min}$ (the denaturant concentration where the folding and unfolding rates are equal) is not the same as $[D]_m$ in general.

Here, $[D]_m$ is the thermodynamic midpoint which can also be infered as the point where the denaturant concentration in which the protein is 50% folded and 50% unfolded.

These two concentrations are equal only when $m_u = |m_f|$.

Equilibrium thermodynamics from kinetics

The equilibrium constant for folding ($K_{\text{eq}} = [\text{F}]/[\text{U}]$) is given by the equation :

$$K_{\text{eq}}([D]) = \frac{k_{\text{f}}([D])}{k_{\text{u}}([D])} = \frac{k_{\text{f}}^{\text{w}}}{k_{\text{u}}^{\text{w}}} \exp\left(\frac{(m_f - m_u)[D]}{RT}\right) \tag{6}$$

The free energy of unfolding and the denaturation midpoint are given by the set of equations :

$$\Delta G_U([D]) = RT \ln K_{\text{eq}}([D]) = \Delta G_U(0) + (m_f - m_u)[D] \tag{7}$$

$$[D]_m = \frac{\Delta G_U(0)}{m_u - m_f} = \frac{RT \ln(k_{\text{f}}^{\text{w}}/k_{\text{u}}^{\text{w}})}{m_u - m_f} \tag{8}$$

The fraction unfolded is obtained from the sigmoidal transition:

$$f_U([D]) = \frac{1}{1 + K_{\text{eq}}([D])} \tag{9}$$

ϕ value calculation

Using Eq. (37), the ϕ-value at $[D] = 0\,\text{M}$ is:

$$\phi(0\,\text{M}) = \frac{RT \ln(k_{\text{f}}^{\text{WT}}/k_{\text{f}}^{\text{mut}})}{RT \ln(K_{\text{eq}}^{\text{WT}}/K_{\text{eq}}^{\text{mut}})} \tag{10}$$

At the WT midpoint $[D]_m^{\text{WT}}$, where $K_{\text{eq}}^{\text{WT}}([D]_m) = 1$, this simplifies to:

$$\phi([D]_m) = \frac{\ln\left[k_{\text{f}}^{\text{mut}}([D]_m)/k_{\text{f}}^{\text{WT}}([D]_m)\right]}{\ln K_{\text{eq}}^{\text{mut}}([D]_m)} \tag{11}$$

where all rates are evaluated at $[D] = [D]_m^{\text{WT}}$ using the mutant's own m_f, m_u.

Results

Wild-type parameters

The Wild type protein has $k_{\text{f}}^{\text{w}} = 27\,600\,\text{s}^{-1}$ and $k_{\text{u}}^{\text{w}} = 20\,\text{s}^{-1}$ in water, with $m_f = -2.301$, $m_u = 1.255\,\text{kJ mol}^{-1}\,\text{M}^{-1}$. Other quantities derived from this data are shown in Table 1.

Table 1: Wild-type thermodynamic and kinetic parameters at 298 K in GuHCl.

Quantity	Value	Units
k_{f}^{w}	27 600	s^{-1}
k_{u}^{w}	20	s^{-1}
m_f	-2.301	$\text{kJ mol}^{-1}\,\text{M}^{-1}$
m_u	$+1.255$	$\text{kJ mol}^{-1}\,\text{M}^{-1}$
$m_{\text{eq}} = m_u - m_f$	3.556	$\text{kJ mol}^{-1}\,\text{M}^{-1}$
$K_{\text{eq}}(0\,\text{M})$	1380	-
$\Delta G_U(0\,\text{M})$	17.91	kJ mol^{-1}
$[D]_m^{\text{WT}}$ (Thermodynamic Min C_m)	5.037	M
$[D]_{\min}$ (Kinetics Min (Chevron Min))	5.459	M

Chevron plots

Figure 1 shows the chevron plots for all 22 mutants along with the wild type which is shown in Black. It's an important observation to be made that all the mutant variants retain the classic V-shape of the plot. The left folding arm and right unfolding arm have different slopes because of the data which had $|m_f| > m_u$ for most of the mutant samples, From our code we got the kinetic minimum to be equal to 5.46 M which is to the *right* of the thermodynamic midpoint at 5.04 M (Eq. 5).

From our Chevron Plot we see that all mutations lead to lower stability than the wild type at the concentration when the Wild type is most stable; Mutations such as Y13F in which Tyrosine that contains a hydroxyl group capable of forming hydrogen bonds within the protein core when replacing it with phenylalanine removes this interaction. The loss of that hydrogen bond likely weakens the folded state which reflects why in the plot we see that at the $[D]$ conc when Wild type is most stable the mutation shows maximum positive deviation

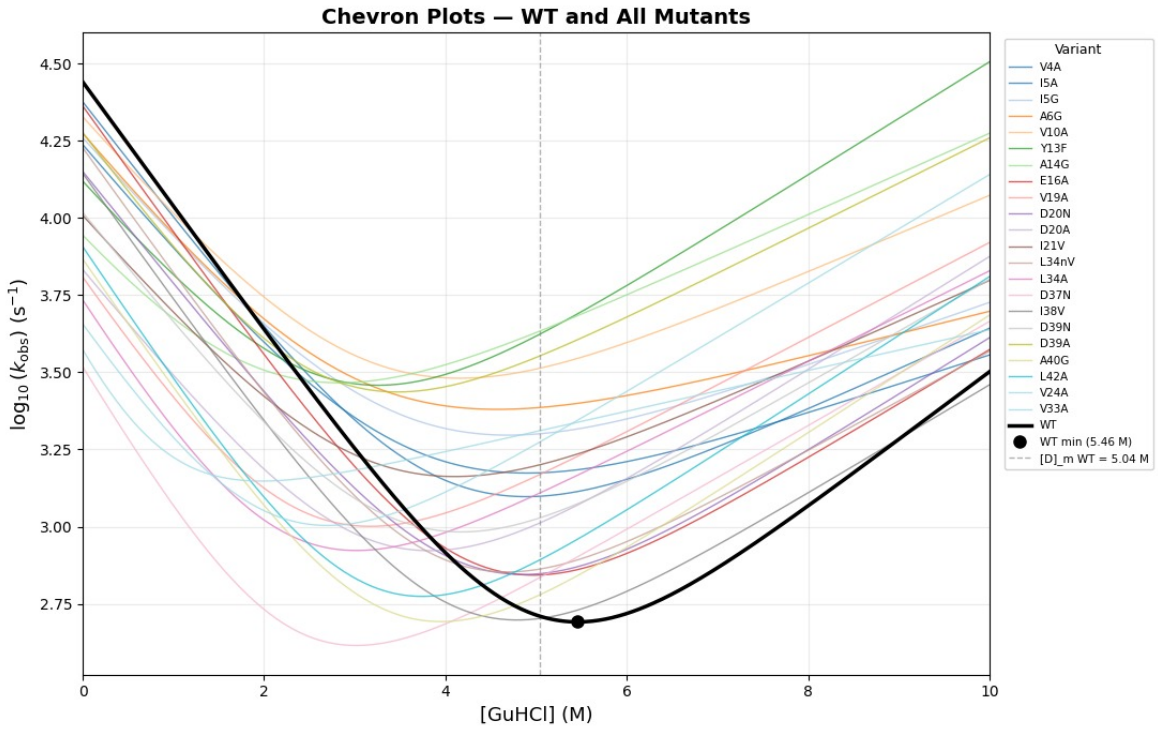


Figure 1: Chevron plots of $\log_{10}(k_{\text{obs}})$ vs. [GuHCl] for the WT (black, thick) and all 22 mutants (colours). The dashed grey vertical line marks the WT thermodynamic midpoint $[D]_m^{\text{WT}} = 5.04$ M; the dotted line and filled circle mark the kinetic minimum at 5.46 M.

ϕ-values at 0 M denaturant

The ϕ-values were calculated at $[D] = 0$ M (Eq. 10) are shown in Figure 2 and tabulated in Table 5. The distribution of the ϕ values ranges from $\phi = 0.06$ (V4A) to $\phi = 0.58$ (D37N). The normal standard cutoff for what contributes to the folding nucleus is 0.5 but we have also marked the Residues with $\phi \geq 0.4$ which are highlighted in orange in the figure as possible folding nucleus candidates.

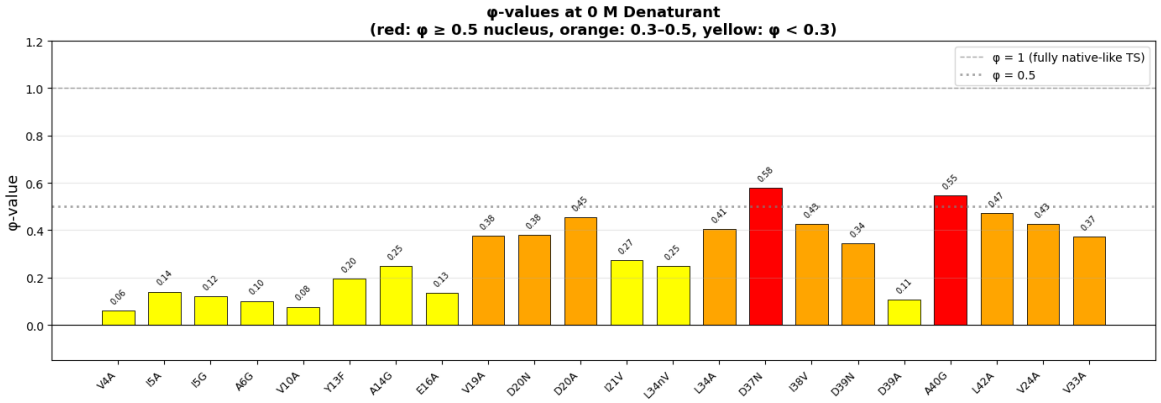


Figure 2: ϕ-values at 0 M GuHCl for all 22 mutants. Bars are colour-coded: **red** ($\phi \geq 0.5$, strong nucleus); **orange** ($0.3 \leq \phi < 0.5$, partial involvement); **yellow** ($\phi < 0.3$, little TS involvement). Dashed line at $\phi = 1$ (fully native-like TS). Values are annotated above each bar.

Folding nucleus identification

Residues with $\phi \geq 0.5$ can be considered to be part of the folding nucleus. This is because a ϕ value greater than 0.5 indicates that in the transition state more than half of the final folded state interactions at that residue are already formed. In other words we can say that part of the protein is closer to the folded state than the unfolded state, which is a key sign for residues that belong to the folding nucleus.

Table 2: Probable folding nucleus residues when ($\phi \geq 0.40$ at 0 M GuHCl), ranked by ϕ-value. $\Delta\Delta G^{\ddagger}$ and in kJ mol^{-1}.

Mutation	Residue	k_f^{mut} (s^{-1})	$\Delta\Delta G^{\ddagger}$	$\Delta\Delta G_{eq}$	ϕ
D37N	37	3 200	5.34	9.20	**0.580**
A40G	40	7 300	3.30	6.02	**0.548**
L42A	42	8 000	3.07	6.50	0.472
D20A	20	6 700	3.51	7.73	0.454
I38V	38	13 900	1.70	3.97	0.428
V24A	24	4 300	4.61	10.76	0.428
L34A	34	5 200	4.14	10.19	0.406

To do further analysis, we saw all the mutants that had phi greater than 0.4 (chosen arbitrarily). From the Data generated we see that the nucleus is mostly in the C-terminal half of the 45-residue chain (residues 34, 37, 38, 40, 42) which is consistent with a native-like folding transition state that nucleates from this region. D37 and A40 show the highest ϕ-values (≥ 0.5) thus one can make the clear biological inference that these residues form the structural core of the transition state.

ϕ-values at the WT denaturation midpoint

At the Wild Type midpoint Denaturant Concentration which was calculated to be ($[D]_m^{\mathrm{WT}} = 5.037\,\mathrm{M}$), the rates are: $k_{\mathrm{f}}^{\mathrm{WT}}([D]_m) = k_{\mathrm{u}}^{\mathrm{WT}}([D]_m) = 256.58\,\mathrm{s}^{-1}$, confirming the fact that $K_{\mathrm{eq}}^{\mathrm{WT}}([D]_m) = 1$ exactly, which is as per its definition.

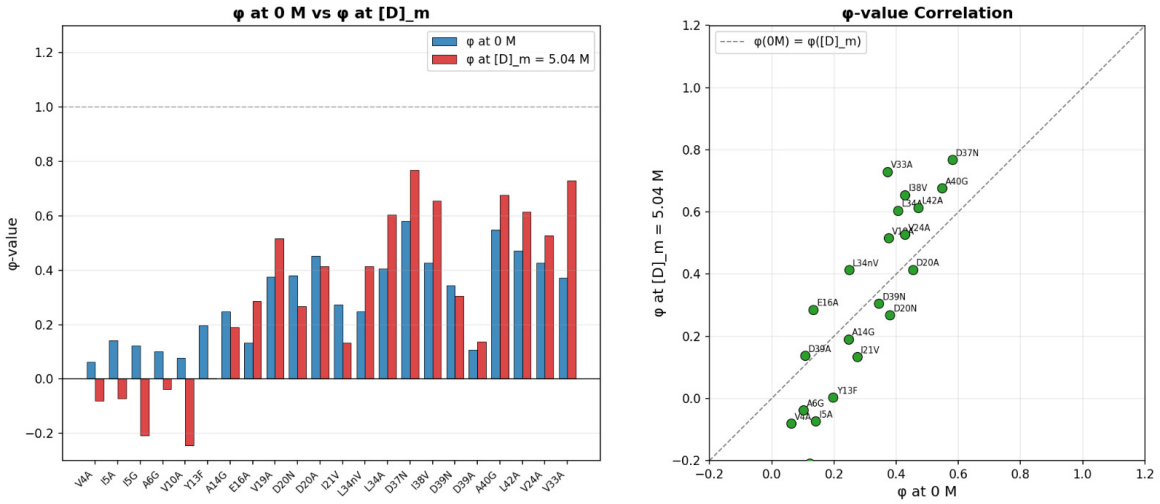


Figure 3: *Left*: Grouped bar chart comparing $\phi(0\,\mathrm{M})$ (blue) and $\phi([D]_m^{\mathrm{WT}})$ (red) for each mutant. *Right*: Scatter plot correlation; the dashed diagonal is the identity line.

Figure 3 and Table 5 Gives us greater insights into some of the features that are present when we compare the ϕ values at Midpoint concentration (C_m) and when it was previously calculated at $0M$ Denaturant concentration. These Include :

(i) **Nucleus residues increase.** The mutants that have a high ϕ value , some of the examples include (D37N, A40G, L42A, L34A, V33A, I38V) that are taken from our data show *larger* ϕ at the midpoint compared to the residues that had a lower ϕ values under $0M$ denaturant concentration due to the fact that folding rates at higher concentrations of $[D]$ are more selectively perturbed in native like transition states because the it is already very similar to the original structure. So mutations at these key residues affect folding more strongly at higher $[D]$, leading to larger changes in folding rates and hence higher ϕ-values at the midpoint.

(ii) **Low-ϕ mutants can become negative.** We made a very interesting observation that mutants such as V4A, I5A, I5G, A6G, and V10A end up acquiring a small *negative* ϕ-values at the midpoint. This took us by surprise in the beginning but upon further reading we found that this is *physically real* and not just a numerical artifact , Due to the fact that these mutants have shallower folding m values that were seen ($|m_f^{\mathrm{mut}}| < |m_f^{\mathrm{WT}}|$), so at the midpoint concentration of 5.04 M their folding rate *exceeds* the Wild type folding rate despite being less stable overall.

(iii) **Consistency.** The residues that were previously found to be involved in the nucleus like D37N, A40G, L34A, V33A, I38V still have high values of ϕ at the midpoint concentration

Equilibrium denaturation curves

2-state Equilibrium reaction was simulated using the kinetically determined $K_{eq}[D]$ (refer to equation (6)) which was first calculated followed by plotting the fraction of the unfolded state using equation (9) . Figure 4 shows the unfolded state fraction of the wild type along with all the mutants

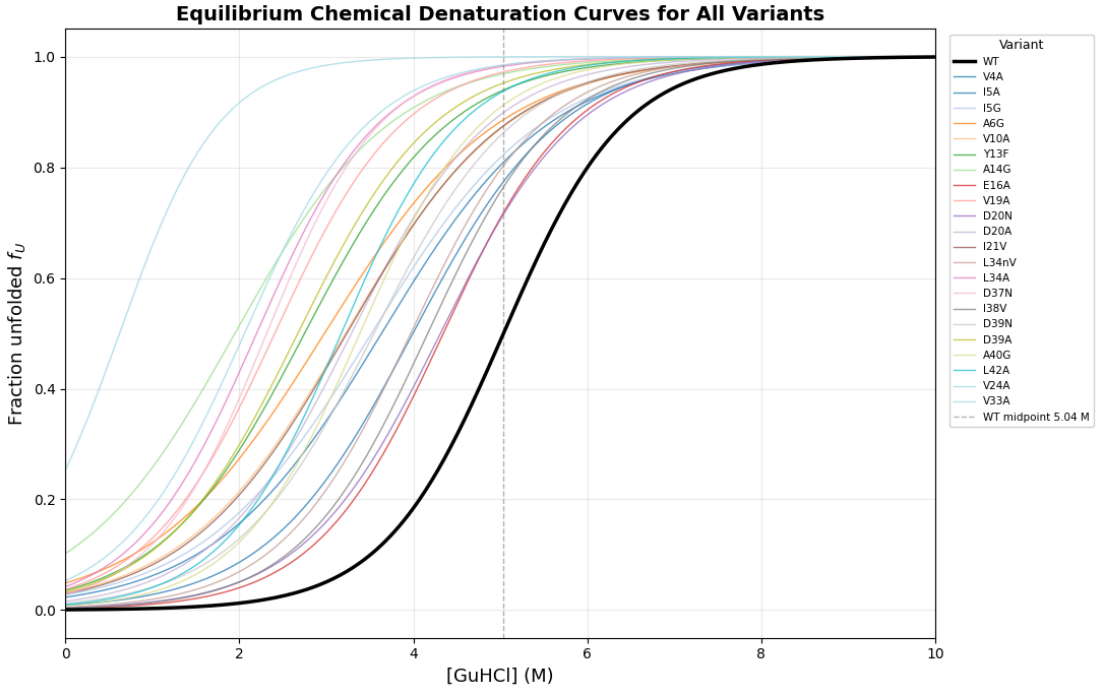


Figure 4: Simulated equilibrium chemical denaturation curves for WT (black, thick) and all 22 mutants (colours). The dashed vertical line marks the WT midpoint at 5.04 M; V33A (unfolds at ≈ 0.62 M) is the most destabilised variant; E16A and D20N are the most stable mutants.

Figure 5 shows the individual stabilities and midpoint Concentrations. The wild type is the most stable variant ($\Delta G_U = 17.91\,\text{kJ mol}^{-1}$, $[D]_m = 5.04\,\text{M}$). V33A from the figure is by far the most destabilised mutant ($\Delta G_U = 2.68\,\text{kJ mol}^{-1}$), The stability trend correlates with ϕ , we see that the low-ϕ residues (N-terminal) when mutated tend to destabilize the protein more than expected from the transition state alone which reflects their importance in the native state contacts not represented in the transition state of the system.

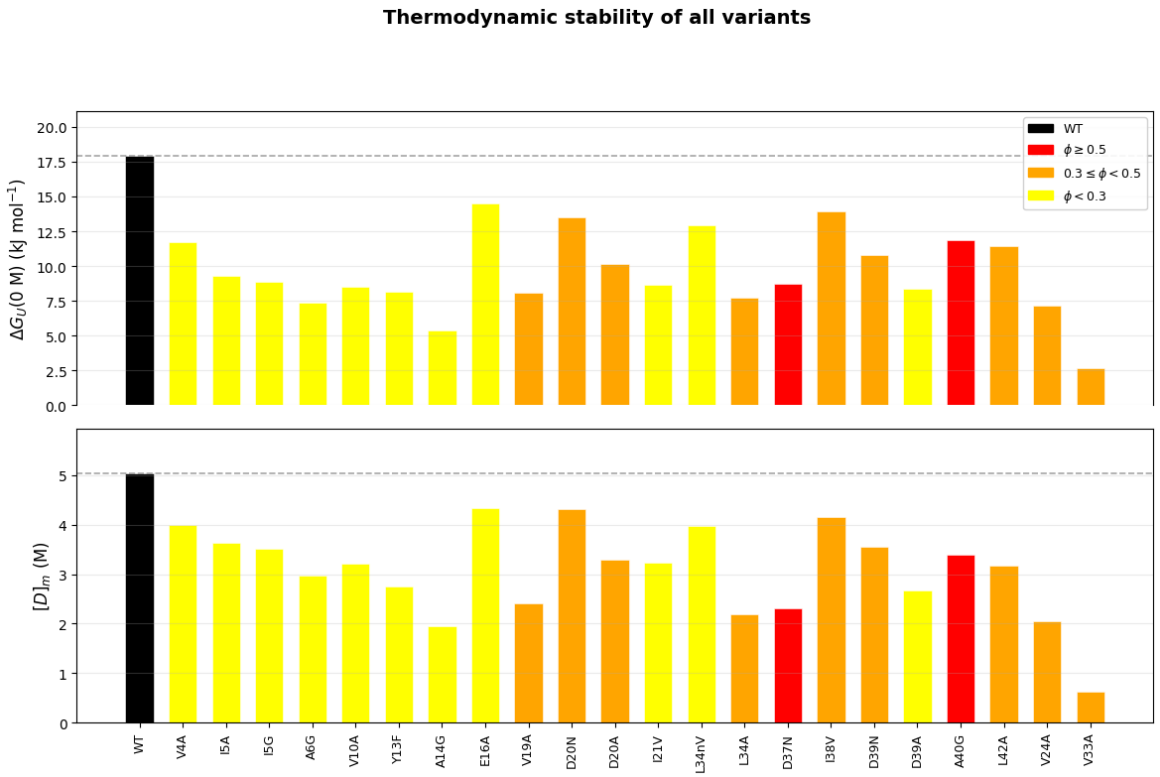


Figure 5: *Top*: $\Delta G_U(0\text{M}) = RT\ln(k_f/k_u)$ for WT and all mutants. *Bottom*: Denaturation midpoint $[D]_m$. Bars are coloured by ϕ-value class (same scheme as Fig. 2). Dashed lines indicate WT values.

Fig : 6 shows the simulation results for each of the mutants and the wild type along with the gibbs free energy associated with each of them.

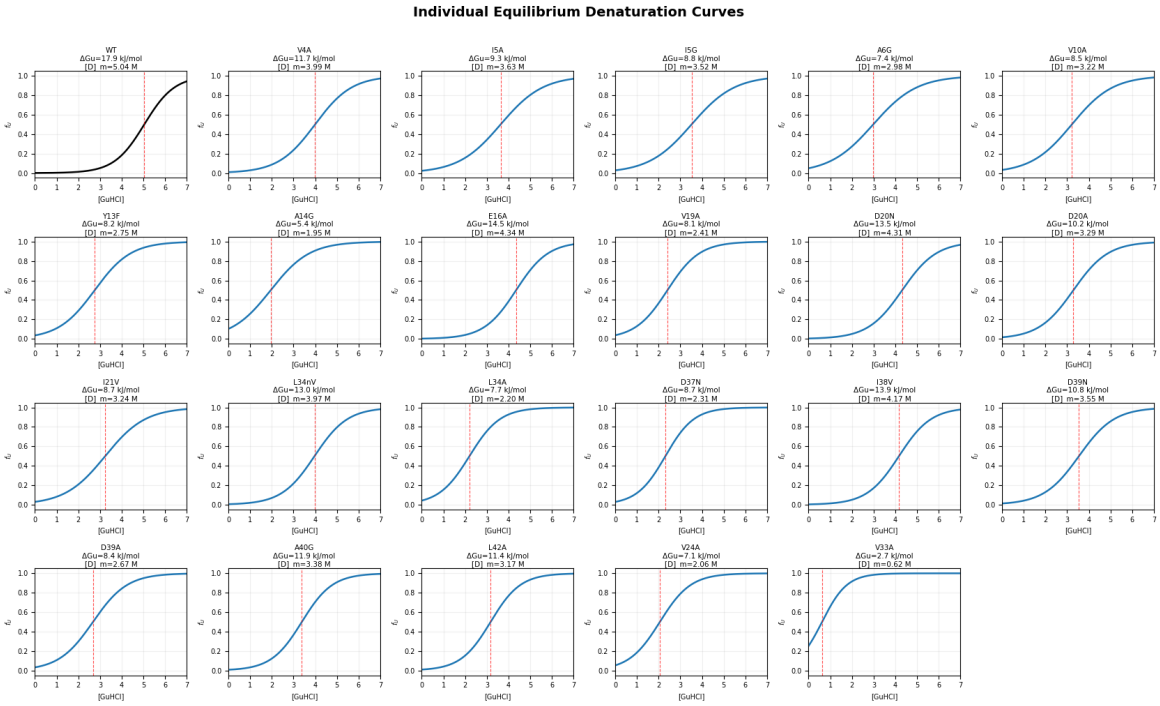


Figure 6: Chemical Denaturation Curve (Individual)

Extension to a Three-State System

For all the derivations and notations used in our 3 state model system we followed the paper [2]

Experimental strategy

The two state protein engineering approach can be extended further to a three state system by incorporating an intermediate state (I) and two transition states (TS_1, TS_2). The steps involved to achieve this

(1) First Introduce a mutation and measure the change in overall stability:

$$\Delta\Delta G_{UF}$$

(2) Then measure the kinetic changes (folding/unfolding rates) from experiments such as chevron plots.

(3) Define the ϕ-values for each transition state (3 state model will have 2 ϕ) values corresponding to the $TS1$ and $TS2$:

$$\phi_1 = \frac{\Delta\Delta G_1^{\ddagger}}{\Delta\Delta G_{UF}}, \qquad \phi_2 = \frac{\Delta\Delta G_2^{\ddagger}}{\Delta\Delta G_{UF}} \tag{12}$$

Idea: Mutations change the kinetics and the stability and from these changes we can infer when different parts of the protein fold along the pathway $U \to I \to F$.

Mechanism and notation

Consider a three-state mechanism with a populated intermediate I:

$$\mathrm{U} \underset{k_{u1}}{\overset{k_{f1}}{\rightleftharpoons}} \mathrm{I} \underset{k_{u2}}{\overset{k_{f2}}{\rightleftharpoons}} \mathrm{F} \tag{13}$$

with denaturant-dependent rates:

$$\begin{aligned} k_{f1}([D]) &= k_{f1}^{w}\, e^{m_{f1}[D]/RT}, & k_{u1}([D]) &= k_{u1}^{w}\, e^{m_{u1}[D]/RT} \\ k_{f2}([D]) &= k_{f2}^{w}\, e^{m_{f2}[D]/RT}, & k_{u2}([D]) &= k_{u2}^{w}\, e^{m_{u2}[D]/RT} \end{aligned} \tag{14}$$

Kinetics:

The rate equations are:

$$\frac{d[U]}{dt} = -k_{f1}[U] + k_{u1}[I] \tag{15}$$

$$\frac{d[I]}{dt} = k_{f1}[U] - (k_{u1} + k_{f2})[I] + k_{u2}[F] \tag{16}$$

$$\frac{d[F]}{dt} = k_{f2}[I] - k_{u2}[F] \tag{17}$$

In matrix form:

$$\frac{d}{dt}\begin{pmatrix}[U]\\[I]\\[F]\end{pmatrix} = \begin{pmatrix}-k_{f1} & k_{u1} & 0\\ k_{f1} & -(k_{u1}+k_{f2}) & k_{u2}\\ 0 & k_{f2} & -k_{u2}\end{pmatrix}\begin{pmatrix}[U]\\[I]\\[F]\end{pmatrix} \tag{18}$$

To understand how these linear set of equations changes over time, we analyse the above rate matrix by finding its corresponding eigenvalues. This helps us identify the timescales of the system and develop the final set of rate equations for each of the 3 states Using the concepts from our non linear dynamics .

In general, the solution to these coupled differential equations can be written as a sum of exponential terms. Each eigenvalue corresponds to one such exponential decay, and therefore represents a kinetic phase that can be observed experimentally.

For a three-state system, we obtain three eigenvalues. One of them is zero, which corresponds to the equilibrium state (no further change in concentrations). The other two eigenvalues are non-zero and represent the other observed states of the system.

To decouple the system, we first find the eigenvalues (λ) of the rate matrix, K, by solving the characteristic equation $\det(K - \lambda I) = 0$:

$$\begin{vmatrix} -k_{f1} - \lambda & k_{u1} & 0 \\ k_{f1} & -(k_{u1} + k_{f2}) - \lambda & k_{u2} \\ 0 & k_{f2} & -k_{u2} - \lambda \end{vmatrix} = 0 \tag{19}$$

This gives us a cubic equation in λ:

$$\lambda \left(\lambda^2 + (k_{f1} + k_{u1} + k_{f2} + k_{u2})\lambda + (k_{f1}k_{f2} + k_{f1}k_{u2} + k_{u1}k_{u2}) \right) = 0 \tag{20}$$

This equation has three roots; the first one corresponds to the equilibrium state where the net rate of change is zero:

$$\lambda_0 = 0 \tag{21}$$

The remaining two roots correspond to the observable kinetic phases. To simplify the expressions, we can make the following substitutions:

$$b = k_{f1} + k_{u1} + k_{f2} + k_{u2} \tag{22}$$

$$c = k_{f1}k_{f2} + k_{f1}k_{u2} + k_{u1}k_{u2} \tag{23}$$

Using the quadratic formula, the two kinetic eigenvalues are:

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4c}}{2} \tag{24}$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4c}}{2} \tag{25}$$

Next, we get the corresponding eigenvectors, $\vec{v}_i$, by solving $(K - \lambda_i I)\vec{v}_i = 0$.

For the equilibrium eigenvalue ($\lambda_0 = 0$), the eigenvector $\vec{v}_0$ represents the steady-state fractional populations. By defining the equilibrium constants for each step as $K_1 = k_{f1}/k_{u1}$ and $K_2 = k_{f2}/k_{u2}$, and since total mass is constant ($[U] + [I] + [F] = 1$), we obtain:

$$\vec{v}_0 = \frac{1}{1 + K_1 + K_1K_2} \begin{pmatrix} 1 \\ K_1 \\ K_1K_2 \end{pmatrix} = \begin{pmatrix} [U]_{eq} \\ [I]_{eq} \\ [F]_{eq} \end{pmatrix} \tag{26}$$

The eigenvalues (λ_1 and λ_2) tell us how fast different parts of the system change over time, and the eigenvectors show how much each state contributes to those changes. Their eigenvectors can be

written as:

$$\vec{v}_{1,2} = \begin{pmatrix} 1 \\ \frac{k_{f1}+\lambda_{1,2}}{k_{u1}} \\ \frac{k_{f2}(k_{f1}+\lambda_{1,2})}{k_{u1}(k_{u2}+\lambda_{1,2})} \end{pmatrix} \tag{27}$$

The general solution of the system is a linear combination of these eigenvectors:

$$\begin{pmatrix} [U](t) \\ [I](t) \\ [F](t) \end{pmatrix} = c_0\vec{v}_0 e^{\lambda_0 t} + c_1\vec{v}_1 e^{\lambda_1 t} + c_2\vec{v}_2 e^{\lambda_2 t} \tag{28}$$

Since $\lambda_0 = 0$ and the equilibrium populations are given by $\vec{v}_0$ (with $c_0 = 1$), the fully decoupled time-dependent equations for the three states are:

$$[U](t) = [U]_{eq} + c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} \tag{29}$$

$$[I](t) = [I]_{eq} + c_1 \left(\frac{k_{f1}+\lambda_1}{k_{u1}}\right) e^{\lambda_1 t} + c_2 \left(\frac{k_{f1}+\lambda_2}{k_{u1}}\right) e^{\lambda_2 t} \tag{30}$$

$$[F](t) = [F]_{eq} + c_1 \left(\frac{k_{f2}(k_{f1}+\lambda_1)}{k_{u1}(k_{u2}+\lambda_1)}\right) e^{\lambda_1 t} + c_2 \left(\frac{k_{f2}(k_{f1}+\lambda_2)}{k_{u1}(k_{u2}+\lambda_2)}\right) e^{\lambda_2 t} \tag{31}$$

The amplitudes c_1 and c_2 depend strictly on the initial boundary conditions (e.g., $[U](0) = 1$, $[I](0) = 0$, and $[F](0) = 0$ for a standard folding experiment from a fully unfolded state) and can be solved algebraically by evaluating the system at $t = 0$.

Equilibrium Populations

The equilibrium constants for each step are:

$$K_1([D]) = \frac{k_{f1}([D])}{k_{u1}([D])}, \qquad K_2([D]) = \frac{k_{f2}([D])}{k_{u2}([D])} \tag{32}$$

With partition function $Z = 1 + K_1 + K_1K_2$, the fractional populations are:

$$P_U = \frac{1}{Z}, \qquad P_I = \frac{K_1}{Z}, \qquad P_F = \frac{K_1K_2}{Z} \tag{33}$$

The overall stability is $\Delta G_{UF} = -RT\ln(K_1K_2)$. When I accumulates, the equilibrium unfolding curve becomes **biphasic**, with a a plateau in the transition region which is a key experimental result of a three state system.

Phi-Value Analysis for Two Transition States

In the two-state case, a single ϕ value measures how much of the total destabilisation $\Delta\Delta G_{UF}$ is felt at the unique transition state. For three states, we define ϕ_1 and ϕ_2 with respect to TS_1 and TS_2 respectively, using the same total stability change as the reference:

$$\Delta\Delta G_{UF} = -RT \ln\left(\frac{K_1^{WT}\, K_2^{WT}}{K_1^{mut}\, K_2^{mut}}\right) \tag{34}$$

$$\Delta\Delta G_1^{\ddagger} = -RT \ln\left(\frac{k_{f1}^{mut}}{k_{f1}^{WT}}\right) \tag{35}$$

$$\Delta\Delta G_2^{\ddagger} = -RT \ln\left(\frac{K_1^{WT}}{K_1^{mut}}\right) - RT \ln\left(\frac{k_{f2}^{mut}}{k_{f2}^{WT}}\right) \tag{36}$$

Note that $\Delta\Delta G_2^{\ddagger}$ is measured *relative to the unfolded state* U, not relative to I. This is essential: if I is itself destabilised by the mutation, that contribution must be included in the barrier height for TS_2. The ϕ-values are then:

$$\phi_1 = \frac{\Delta\Delta G_1^{\ddagger}}{\Delta\Delta G_{UF}}, \qquad \phi_2 = \frac{\Delta\Delta G_2^{\ddagger}}{\Delta\Delta G_{UF}} \tag{37}$$

Physical interpretation

- $\phi_1 \approx 1$: the residue's contacts are fully formed at TS_1 : it is part of the early folding nucleus.
- $\phi_1 \approx 0$: the residue is disordered at TS_1; it may fold in the second step.
- $\phi_2 \approx 1$: the residue is structured at TS_2 - it consolidates during the second folding event.
- $\phi_1 \approx 0$, $\phi_2 \approx 0$: the residue forms contacts only after both transition states (i.e. only in the final native state).
- $\phi_1 \approx 1$, $\phi_2 \approx 1$: the residue is like the final folded structure at *both* transition states : a core nucleus that drives the entire folding reaction.

Parameters and Numerical Setup

The three-state WT parameters are constructed by splitting the real Prot1.dat WT kinetics ($k_{f,WT} = 27600\,\mathrm{s}^{-1}$, $k_{u,WT} = 20\,\mathrm{s}^{-1}$, $m_f = -2.30\,\mathrm{kJ\,mol^{-1}M^{-1}}$, $m_u = 1.26\,\mathrm{kJ\,mol^{-1}M^{-1}}$) across two barriers.

Table 3: Three-state WT parameters used in this analysis (GuHCl, 298 K).

Parameter	Step 1 ($U \to I$)	Step 2 ($I \to F$)
$k_{f,0}$ (s^{-1})	41400	2760
$k_{u,0}$ (s^{-1})	6000	40
$K_0 = k_f/k_u$	6.90	69.0
m_f ($\mathrm{kJ\,mol^{-1}M^{-1}}$)	-0.805	-1.496
m_u ($\mathrm{kJ\,mol^{-1}M^{-1}}$)	$+0.251$	$+1.004$

The overall stability $\Delta G_{UF} = -RT\ln(K_1K_2) = -15.28\,\mathrm{kJ\,mol^{-1}}$, consistent with the two-state WT ($-17.91\,\mathrm{kJ\,mol^{-1}}$) to within the uncertainty of the m-value splitting.

Two representative mutants are simulated to demonstrate the limiting ϕ-value cases:

- **Mut-A (early nucleus):** k_{f1} reduced 20-fold, k_{f2} nearly unchanged. Expected: $\phi_1 \approx 1$, $\phi_2 \approx 1$.
- **Mut-B (late nucleus):** k_{f1} nearly unchanged, k_{f2} reduced 25-fold. Expected: $\phi_1 \approx 0$, $\phi_2 \approx 1$.

Table 4: Computed ϕ-values for the two representative mutants.

Mutant	$\Delta\Delta G_{UF}$ (kJ mol^{-1})	ϕ_1	ϕ_2
Mut-A (early nucleus)	8.45	0.878	0.947
Mut-B (late nucleus)	8.71	0.030	0.973

Results and Discussion

Figure 7 shows the three key plots for the three-state system.

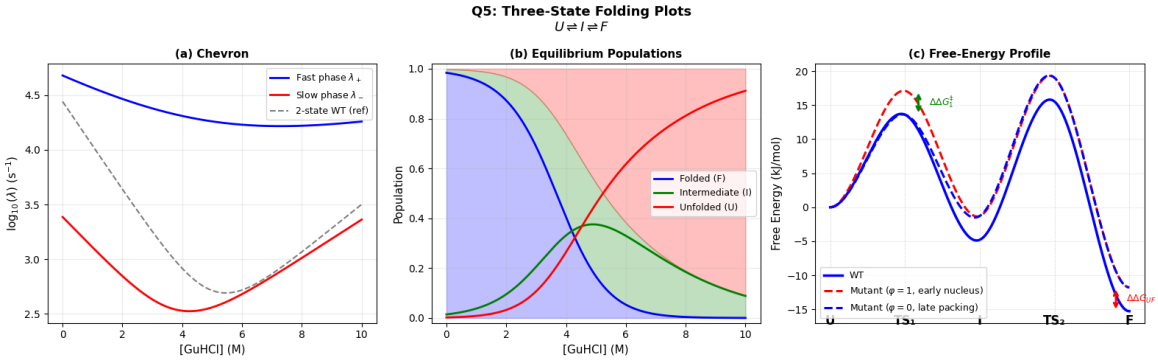


Figure 7: Three-state folding analysis. **(a)** Chevron plot: the slow phase (red) shows a tilt in the folding arm, deviating from the linear two-state WT reference (dashed), which is the kinetic signature of an on-pathway intermediate. **(b)** Equilibrium populations: I accumulates transiently near 1–3 M GuHCl, producing a biphasic denaturation curve. **(c)** Free-energy profile: the $\phi = 1$ mutant (red dashed) raises both barriers equally with $\Delta\Delta G_{UF}$, whereas the $\phi = 0$ mutant (orange dashed) leaves the barriers unchanged while shifting only the ground states.

Phi-Value Interpretation

Both mutants yield $\phi_2 \approx 0.97$, indicating that TS_2 is highly native-like and the second folding step involves significant structure formation in both cases. The key distinction is in ϕ_1:

- Mut-A ($\phi_1 = 0.878$): the first barrier is strongly perturbed : the residue's contacts are mostly formed already at TS_1. This residue is part of the early folding nucleus that stabilises the intermediate.
- Mut-B ($\phi_1 = 0.030$): the first barrier is essentially unaffected : the residue does not contribute to TS_1 structure. It acquires its native contacts only during the second folding step, i.e. the transition from I to F.

Advantage over Two-State Analysis

The two-state ϕ-value takes contributions from both the transition states into a single number.The three state analysis *resolves* these contributions, allowing a step by step map of structure formation: which contacts stabilise the intermediate (probed by ϕ_1) and which drive the final consolidation (probed by ϕ_2). This helps us get more thermodynamic/Kinetic data for proteins where the intermediate is structurally characterized by NMR , as it allows direct comparison between the kinetically and structurally inferred transition state ensembles.

Conclusions

Overall this assignment taught us to analyze systems that have the protein engineering approach through chevron plots and thermodynamic/Kinetic equations . We also extended the 2 state model into a 3 state and derived the expressions for the 2 chevron plots

(1) We saw that the chevron plots confirmed the two state folding kinetics for all variants. The wild type kinetic minimum lies at 5.46 M which is to the right of the thermodynamic midpoint (5.04 M) because of the fact that in the data file our $|m_f| > m_u$.

(2) ϕ-values at 0 M helped us to identify a C terminal folding nucleus comprising of **D37, A40, L42,, I38**, and **L34**. We also saw 2 mutation with ϕ values > 0.5, D37 ($\phi = 0.58$) and A40 ($\phi = 0.55$) form the structural core of the folding nucleus of our protein .

(3) We saw that at the WT midpoint the nucleus residues show higher ϕ-values which helps us to reinforce the structural interpretation previously gotten in our results . Negative ϕ-values for N-terminal residues are physical which have been explained in the results section and is not caused due to an artifact in summary they arise from the m-value heterogeneity across the mutant set.

(4) Equilibrium denaturation curves simulated from kinetic data agree self-consistently with thermodynamic expectations ($f_U([D]_m) = 0.5$ exactly for all variants).

(5) The protein engineering approach extends naturally to three-state systems by: (i) fitting the bi-exponential chevron (eigenvalues of the 3×3 rate matrix), (ii) computing equilibrium populations via K_1 and K_2 and (iii) computing the local ϕ values

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Appendix: ϕ-value Table

Table 5: Complete ϕ-value results for all 22 mutants. Rates in s^{-1}; free energies in kJ mol^{-1}.

Mutant	k_f^{mut}	k_u^{mut}	ΔG_U^{mut}	$[D]_m^{mut}$	$\Delta\Delta G_{eq}$	$\phi(0\,M)$	$\phi([D]_m^{WT})$
V4A	23 600	210	11.70	3.99	6.21	0.062	−0.080
I5A	16 900	400	9.28	3.63	8.64	0.141	−0.073
I5G	17 700	500	8.84	3.52	9.08	0.121	−0.209
A6G	17 900	920	7.35	2.98	10.56	0.102	−0.038
V10A	20 600	670	8.49	3.22	9.43	0.077	−0.245
Y13F	12 700	470	8.17	2.75	9.75	0.197	+0.004
A14G	7 900	900	5.38	1.95	12.53	0.247	+0.190
E16A	23 000	65	14.54	4.34	3.37	0.134	+0.286
V19A	6 200	240	8.06	2.41	9.86	0.375	+0.516
D20N	14 100	60	13.53	4.31	4.39	0.379	+0.268
D20A	6 700	110	10.18	3.29	7.73	0.454	+0.414
I21V	9 900	300	8.66	3.24	9.25	0.275	+0.133
L34nV	16 800	90	12.96	3.97	4.96	0.248	+0.414
L34A	5 200	230	7.73	2.20	10.19	0.406	+0.605
D37N	3 200	95	8.71	2.31	9.20	**0.580**	+0.767
I38V	13 900	50	13.94	4.17	3.97	0.428	+0.655
D39N	10 300	130	10.83	3.55	7.08	0.345	+0.305
D39A	18 300	620	8.39	2.67	9.53	0.107	+0.137
A40G	7 300	60	11.90	3.38	6.02	**0.548**	+0.676
L42A	8 000	80	11.41	3.17	6.50	0.472	+0.614
V24A	4 300	240	7.15	2.06	10.76	0.428	+0.527
V33A	2 800	950	2.68	0.62	15.23	0.372	+0.730